DeCad

Precision and unseen-circuit upgrade

24 October 2026 | Verified baseline: 9985afa | Fresh serialized eighteen-case acceptance

Measurement	Before	Verified after
Clear exact assignments	227/245	228/245 (93.06%)
Native valid physical pins	103	131
Native correct connected pairs	9	10
Native incorrect connected pairs	13	0
Raw Faculty XML exports	7/18	11/18
Unseen automatic components	2/8	8/8
Unseen automatic physical pins	0/19	18/19

All previously correct clear assignments, native physical pins and native connected pairs are retained. The OCR model is unchanged. Automatic recognition remains incomplete; source review is still required.

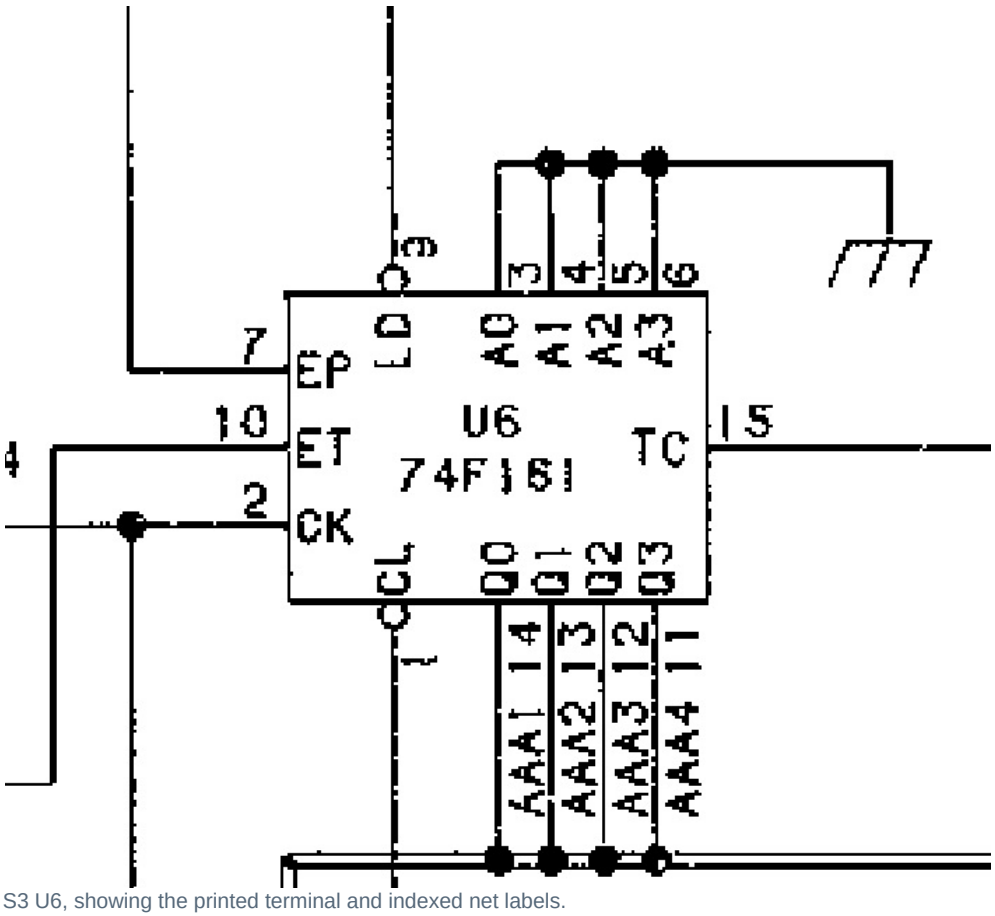
This report separates image accuracy, terminal-number equivalence and faithful export/readback. A passing export does not certify the underlying raster interpretation.

Clear and degraded recognition

Sample	Before	After	Change
S1	54/56	54/56	+0
S2	0/0	0/0	+0
S3	62/69	63/69	+1
S4	69/72	69/72	+0
S5	42/48	42/48	+0
S4-scan300	44/72	44/72	+0
S4-scan200	0/72	0/72	+0
S4-blur300	0/72	0/72	+0

The four clear samples use the existing manually annotated 245-pin reference. S2 exercises component/value recognition and has no annotated pin-to-net target. Scan200 and blur300 remain difficult; unchanged zero scores are shown explicitly.

A missing terminal is added only when a printed package pin, aligned external label and continuous outside conductor agree. Existing assignments are retained. Candidate rules that produced no improvement were reverted.



Source evidence: S3 U6, showing the printed terminal and indexed net labels.

Native stress corpus

Native source	Valid pins	Correct pairs	Wrong pairs
Grove Vision AI V2 / 3b2a49a524e2d50dcf26	29 -> 36	5 -> 5	11 -> 0
a39a7a1fc84f921b27fd	17 -> 17	1 -> 1	0 -> 0
fedf29b44113e135f31f	13 -> 13	0 -> 0	0 -> 0
ESP32 / c4d88e79bc63688e0e7c	4 -> 6	0 -> 0	0 -> 0
XIAO nRF52840 / 37735b083f0b9f56549f	0 -> 0	0 -> 0	1 -> 0
0a236a74292d04d831fa	14 -> 14	0 -> 0	0 -> 0
e9b7597c67252bf7707d	3 -> 11	0 -> 0	0 -> 0
901de51001a98cc2361a	7 -> 9	0 -> 0	0 -> 0
c6bcc506a3200b19e089	12 -> 12	3 -> 3	0 -> 0
CM4Card / c4d7791b4ebdf13c3944	4 -> 13	0 -> 1	1 -> 0

Total valid physical pins: 103 -> 131. Correct pairs: 9 -> 10. Incorrect pairs: 13 -> 0.

Grove ground propagation is blocked at observed capacitor plates. The nRF dotted port labels, including fused alternate-function annotations, are excluded from connector parsing. CM4 internal text no longer spawns U504; the independently captioned U6 owns the outline.

Native coverage is measured against original CAD graphs that can include hidden pin/field information. Several sources remain sparsely recovered. In particular, the nRF guard removes a false interpretation without proving complete device recognition.

Faculty XML and CAD readback

Check	Verified result
Faculty fixture	129/129 nodes; 41 nets; semantic and graph fidelity
Raw corpus XML	11/18 passing; 37 -> 18 zero-pin blockers
KiCad native readback	18/18 cases
OrCAD Capture DSN reopen	18/18 cases
Matched observed assignments	2206/2206 before; 2173/2173 after
Backend tests	412 passed, 6 skipped, 2 warnings in 483.93s (0:08:03)
Frontend tests	18/18 passed

A zero-pin component is quarantined only with absent independent body verification and corroborating alias evidence. Real components with unresolved terminals remain active; they are not hidden to make XML validation pass.

Still blocked	Unresolved component references
S5	R19
S4-scan300	R1, R11, R14
S4-scan200	R11
native-3b2a49a524e2d50dcf26	R32
native-c4d88e79bc63688e0e7c	R15, R2, U4
native-c6bcc506a3200b19e089	R1, R2, R4, R5, R7, R8
native-c4d7791b4ebdf13c3944	R6, R9, U4

Native exports reopen the current observed graph and compare its literal component values and physical pin-to-net assignments. These checks preserve serialization fidelity and do not substitute a library-generated connection for a missing source observation.

Each count uses that version of the observed graph. The corrected inventory removes phantom connector pins and adds supported passive and IC terminals; native source comparison separately confirms retention of previously correct observations.

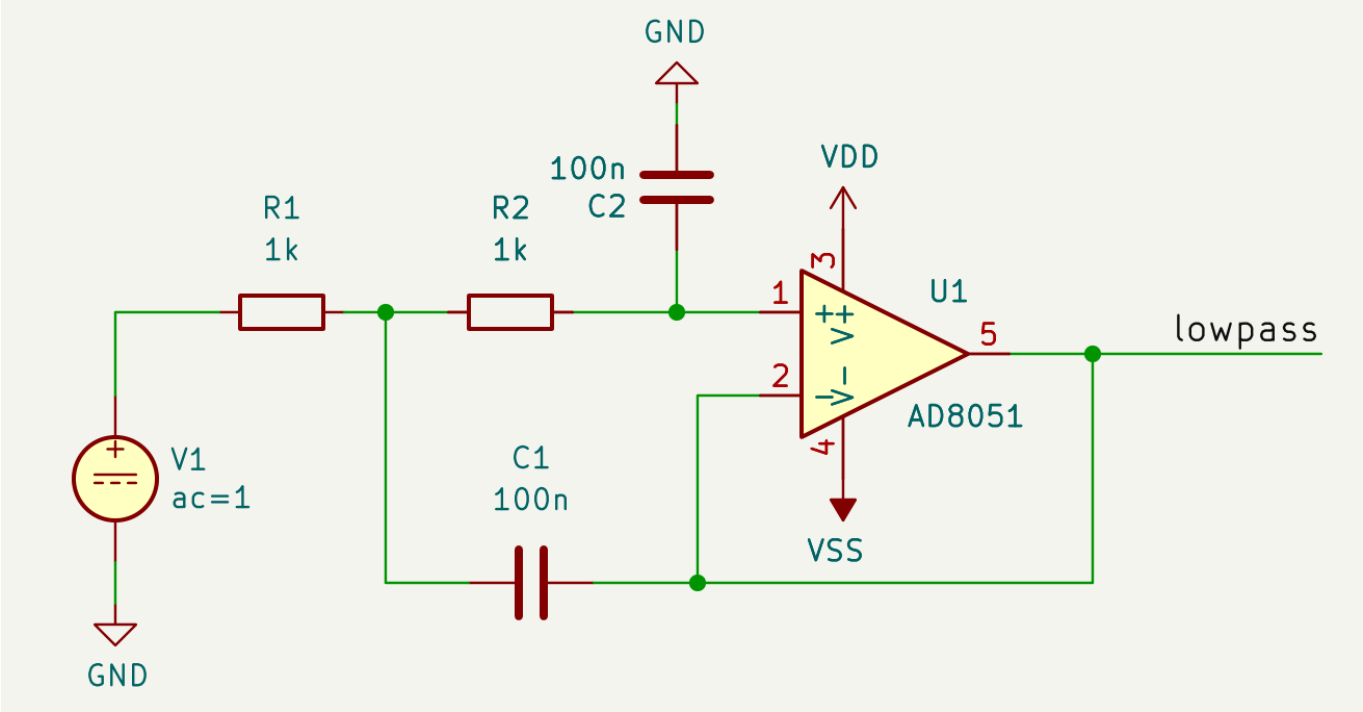
Unseen Sallen-Key: automatic result

Measurement	Literal IDs	R/C/L swap equivalence
Recovered components	8/8	Same components
Recovered physical pins	18/19	Same inventory
Correct connected pairs	5	17
Incorrect connected pairs	12	0
Missing connected pairs	13	1

Unprinted passive terminals use deterministic left/top=1, right/bottom=2 IDs when native symbol orientation is absent. The independently rotated KiCad reference uses the opposite numbering on the four R/C symbols. Equivalence scoring allows only symmetric R/C/L swaps; diodes and sources are excluded. Exports retain the original deterministic IDs.

The unread U1 power terminal remains a source-review item. The independently verified native graph is read only after the raw automatic prediction is saved.

Live API project PrecisionAuto20261024: raster recognition, draft save/reopen, Faculty XML and both native download/readback paths passed without manual correction.



Held-out raster filter source. The supply-source section is also part of the tested image.

Reproduction and remaining limitations

The production profile remains hybrid_max_accuracy. The serialized gate starts a fresh isolated GPU worker for each of eight original/degraded samples and ten native designs. Source fingerprints and loaded CUDA providers are retained in the receipts. The protected model is never trained, replaced or modified.

Raw Faculty XML blockers listed on page 4 require further source-backed terminal recovery. 17 clear pin-to-net errors remain. Low-resolution scans, missing op-amp text and sparsely recovered native designs still require review.

Receipt	Location under evidence/upgrade-20261024
Fresh recognition	final-fresh/gate.json
Raw Faculty XML corpus	xml-corpus/report.json
KiCad and Capture reopen	native-exports/report.json
Unseen raster-only acceptance	unseen-automatic/report.json
Live API and downloads	live-automatic/report.json
Faculty fixture preservation	faculty-roundtrip.json
Exact before/after summary	final-summary.json
Tests	backend-tests-configured.log; frontend-tests.log

Rejected experiments and per-stage checkpoints remain available. Pre-existing unrelated repository changes are preserved outside this task commit. Historical acceptance sections remain in training/TONIGHT_FINAL_ACCEPTANCE.md.

Protected OCR model SHA-256: c5ad0e9ebe43373f1986ed5d6c39be0b73a29cb4dcb257e45ca7f91f41ce07cf

Baseline recovery commits: 9985afa, 871ee61 and b0d6c9b. Delivery receipts record the final implementation commit.

Open review items: clear samples

These remaining mismatches use the existing independent 245-pin manual reference. The expected labels are evaluation data and are never supplied to recognition. Unread entries remain explicit abstentions.

Physical pin	Observed label	Independent expected label
S1.U64.4	WZPLJ6	DD6
S1.U64.13	unread	CC7
S3.U6.14	KAAI	AAA1
S3.U8.14	AAA6	AAA9
S3.U8.11	unread	AAA12
S3.U27.17	unread	AR12
S3.U27.7	RG9	AR3
S3.U28.19	unread	AR10
S4.U31.27	unread	ADDA11
S4.U33.14	unread	ADDA9
S4.U34.9	unread	ADDA7
S5.U36.21	unread	INDA14
S5.U36.2	N00549	INDA12
S5.U49.2	~TT2	INDA8
S5.U49.9	N00619	INBU8
S5.U37.9	unread	INBU4
S5.U50.9	unread	INBU0